THE STRUCTURE OF THE SUN

Background

All stars are similar to ours in their structure and method of producing their energy (fusion of hydrogen). They only vary in their size and temperature, which is indicated by their colour.

Task

Your task is to use the Internet to find useful information that can provide simple answers to the following questions. Several useful sources are listed below.

- Good information - probably the ***best site*** to get information.

  http://www.enchantedlearning.com/subjects/astronomy/sun/sun.shtml

- Good site - lots of information.

  http://www.astronomynotes.com/starsun/s2.htm

You may find others by using ***Metacrawler*** (http://www.metacrawler.com/) and doing a global search on "structure of the sun".

The following sites provide free ***dictionaries*** that may be useful for identifying unknown words.

http://www.thefreedictionary.com/

http://dictionary.reference.com/

Questions

1. (a) How far away is the Sun from Earth (on average)? ______________________________

   (b) On which ***date*** is the Earth closest to the Sun? ______________________________

2. What is the ***temperature*** of the Sun:

   (a) in the ***core***? ____________________

   (b) at the ***surface***? ____________________

   (c) in its atmosphere (called the ***corona***)? ____________________

3. (a) What is the major element found on in the Sun? _________________________

   (b) Name the process used to convert this element into other elements, releasing the huge energy given off by the Sun.

   ____

   (c) During this process (involving Einstein's Equation $E = mc^2$), some of the Sun's mass is converted into pure energy. How much mass does the Sun lose per second in this process?

   ____

4. About how old is the Sun? ______________________________

5. Draw a ***simple labelled diagram*** to show the four major layers of the Sun. Include a brief description of each part.

- Core ______________________________
- Photosphere ______________________________
- Chromosphere ______________________________
- Corona ______________________________

6. (a) What is the ***diameter*** of the Sun in kilometres? ____________________

   (b) How many "Earths" would fit across this diameter? ____________________

   (c) About how many "Earths" could fit in the ***volume*** of the Sun? ____________________

   (d) What percentage (%) of the mass of the entire solar system fits in the Sun? ____________

7. (a) What is a ***sunspot***? ______________________________

   (b) How big can sunspots get (compared to the Earth)? ____________________

   (c) Sunspots appear to occur in a cycle. How long is the cycle? ____________________

   (d) What ***causes*** sunspots? ______________________________

8. The Sun rotates like the Earth but, unlike the Earth, ***the time it takes varies over its surface***.

   (a) What does this tell scientists about the surface of the Sun? ____________________

   (b) Which rotates fastest - the poles of the Sun or the equator? ____________________

9. (a) What is a ***solar flare***? ______________________________

   (b) What is the ***solar wind***? ______________________________

   (c) What is a ***solar prominence***? ______________________________